

Concurrent all-sky gravitational wave transients searches with coherent Wave Burst-2G

joint GR24 and Amaldi16 meeting

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on behalf of cWB-2G team

Outline of the presentation

- **Analysis on public data (15 days of O3b) of simultaneous observations of H,L,V detectors**

Full LVK analysis of O3: [Abbott et al, 2021], Public data available at GWOSC [LINK]

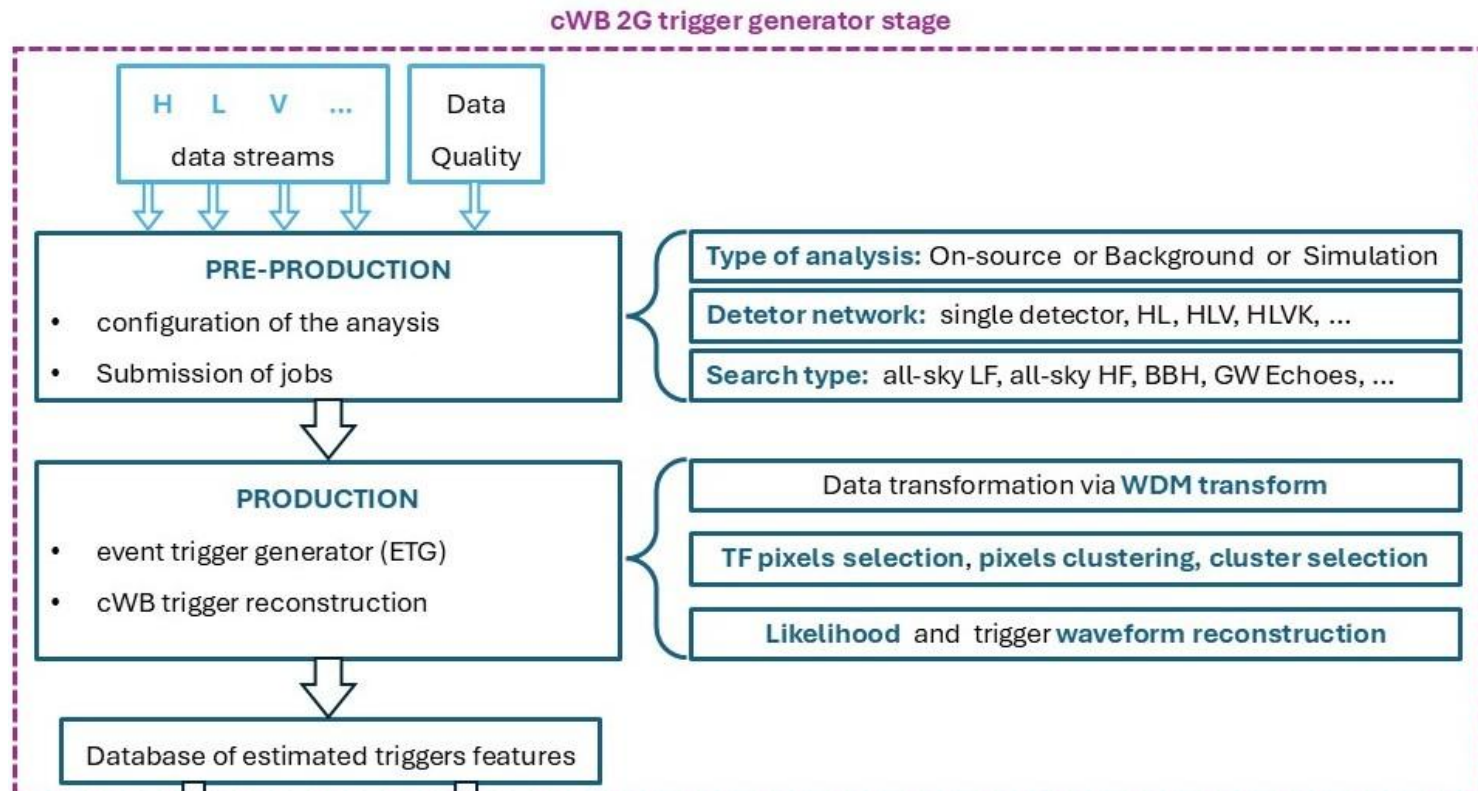
- **Checks of consistency between the empirical background and the Poisson model**
- **Correlation of false alarms between HL and HLV searches**
- **Comparison of search performances HL vs. HLV short-duration burst**

Agnostic Searches and Coherent WaveBurst

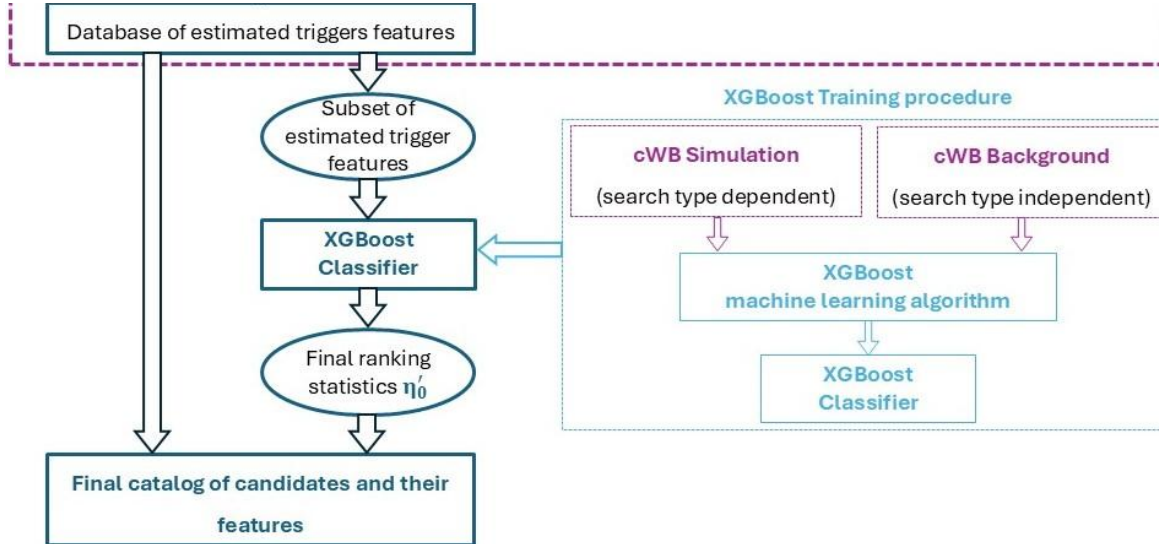
- **Need For Agnostic Searches of gravitational waves transients**
 - Several sources are yet un-modelled or loosely modelled
 - There can be unknown sources of signals
- **Coherent WaveBurst 2G** [Klimenko+2016]
 - Pipeline for **minimally informed** detection and reconstruction of Gravitational Waves signals
 - Based on concept of “Coherent” analysis in a network of detectors (ex: HL, HLV, ...)
- **Machine learning enhanced (cWB-2G-XGBoost)** [Marek+2023]
 - Discrimination between false alarms and signals using a XGBoost model [Chen+2016]
- **This work** [Martini+2025]
 - New ranking statistic
 - Improved triggers' thresholds tuning

Coherent WaveBurst 2G Workflow for O4

[5, Martini et al, in preparation]



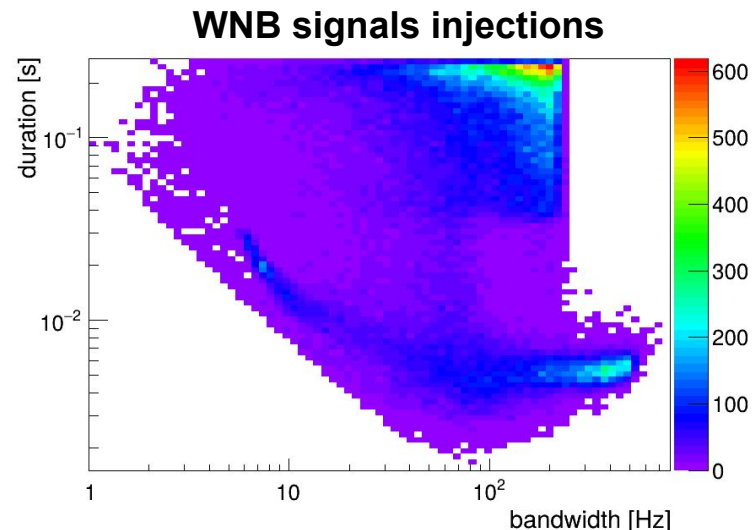
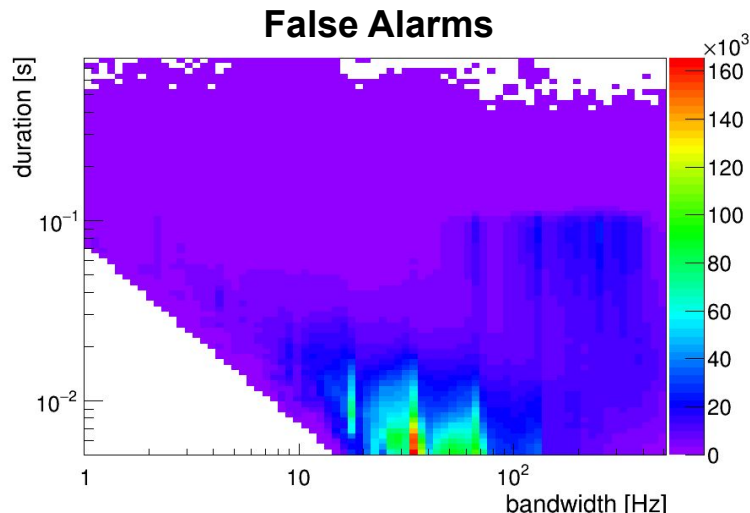
Coherent WaveBurst 2G Workflow for O4



Trigger Features passed to XGBoost:

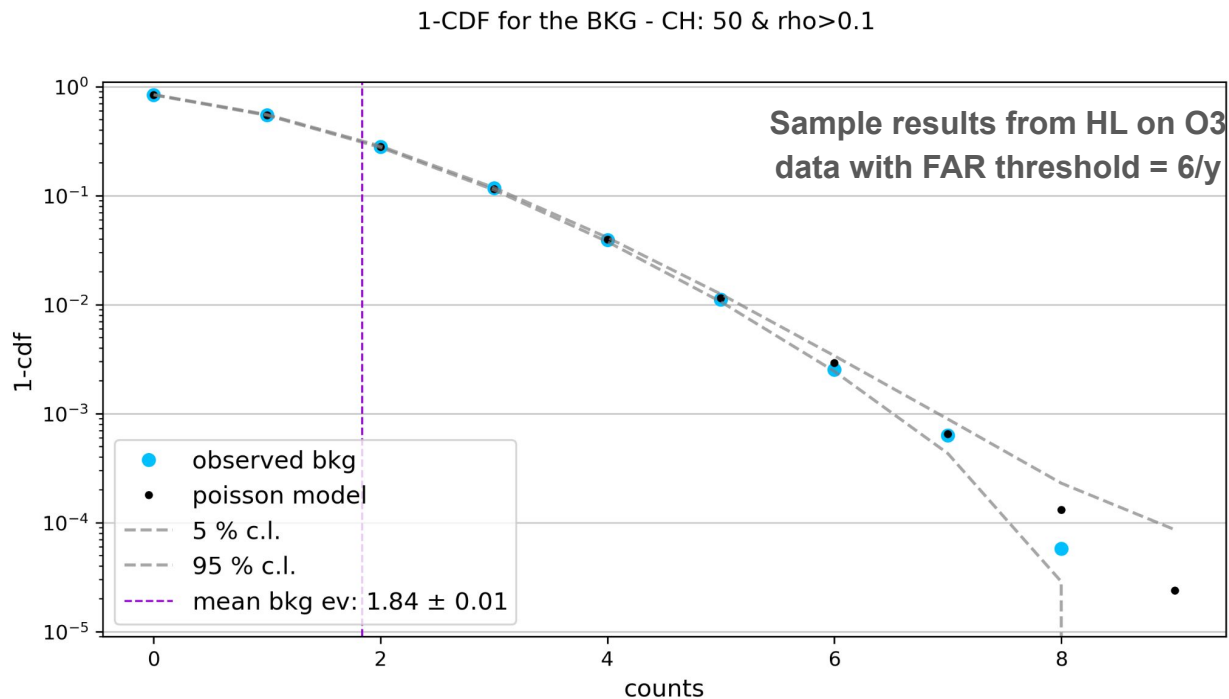
- ρ_0
- norm
- noise
- network correlation coefficient
- χ^2
- mSNR / L
- eCOR / L
- Q_a ,
- Q_p

All-sky short-burst search training procedure - XGBoost



- Background Class
 - The “False Alarm” generated via the time slides
 - 50% for training and 50% for testing
- Signals class:
 - **White Noise Bursts** injections for training
 - More extended morphologies for testing [Marek+2023, Martini+InPreparation]

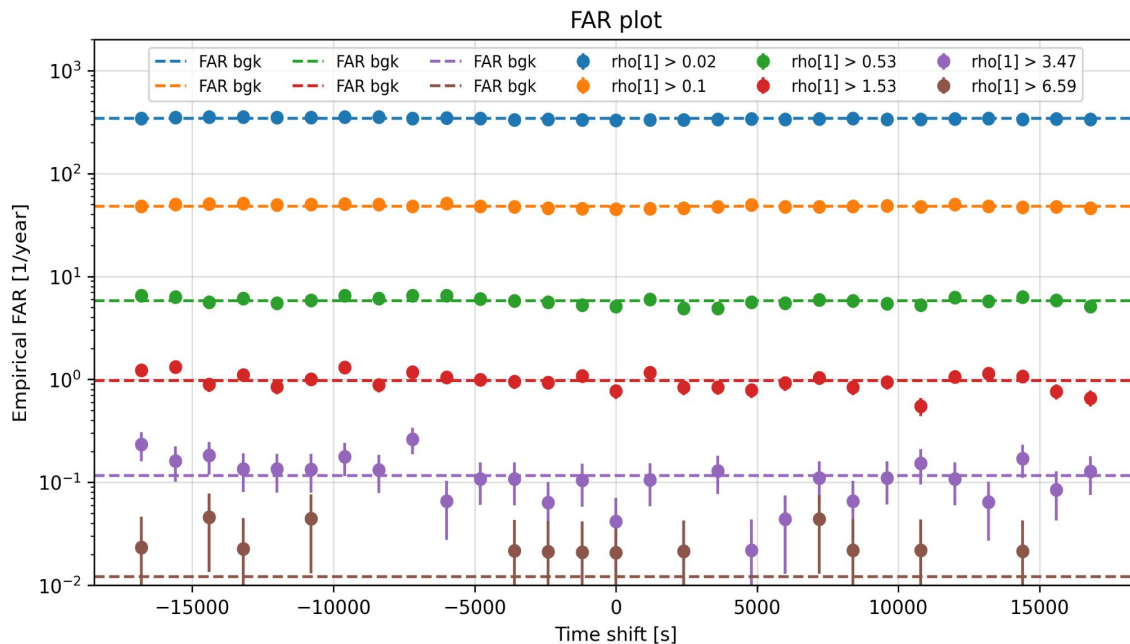
False alarms: Time slides estimates are consistent with Poisson model



1-CDF of BKG counts per time lag in HL short-burst

Distribution of time-slides counts (dots) agree with Poisson model (dots and dashed belt) in a wide FAR range [1/day, /century]

False alarms: Time slides estimates are consistent with Poisson model

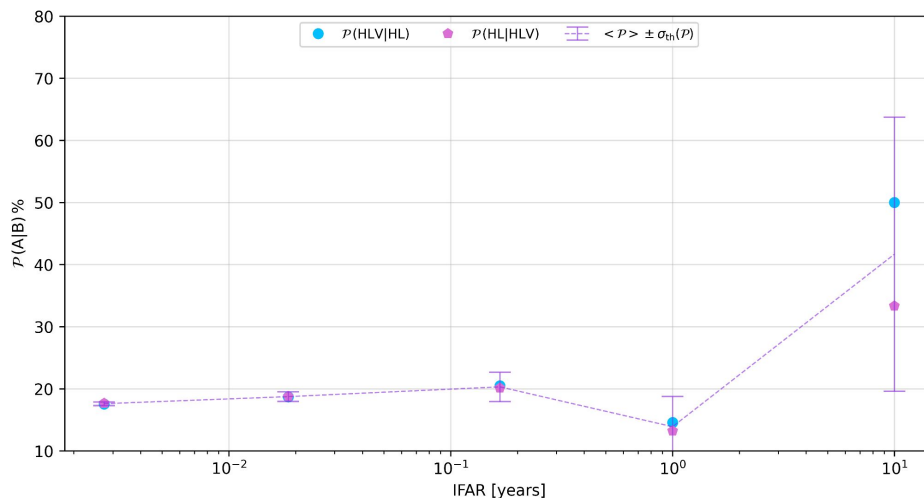
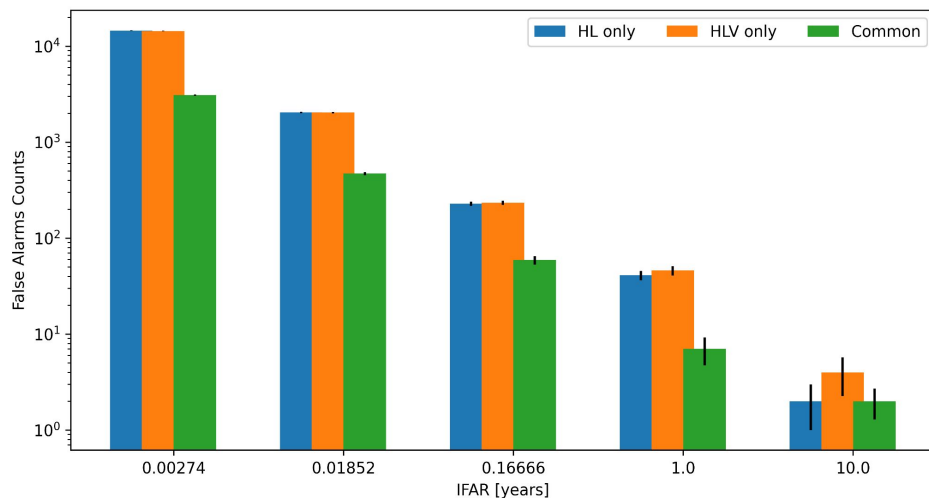


FAR from time-slides for different FAR Thresholds

No correlation of empirical FAR estimates with time shift in a wide FAR range
[1/day - 1/century]

Correlation of false alarms between HL and HLV searches

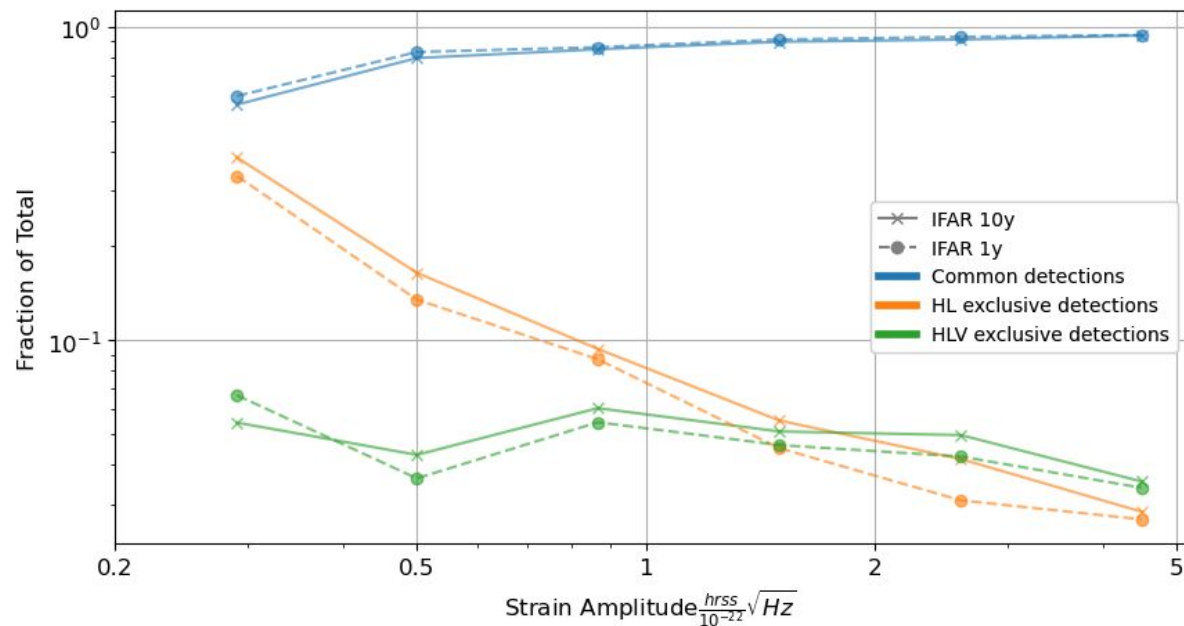
- cWB2G All-sky burst search: **HL vs. HLV**
- **Results based on ~48.2 years of common background livetime**



Common false alarms are only ~ 20%

- **HL and HLV short burst analyses are almost independent searches.**

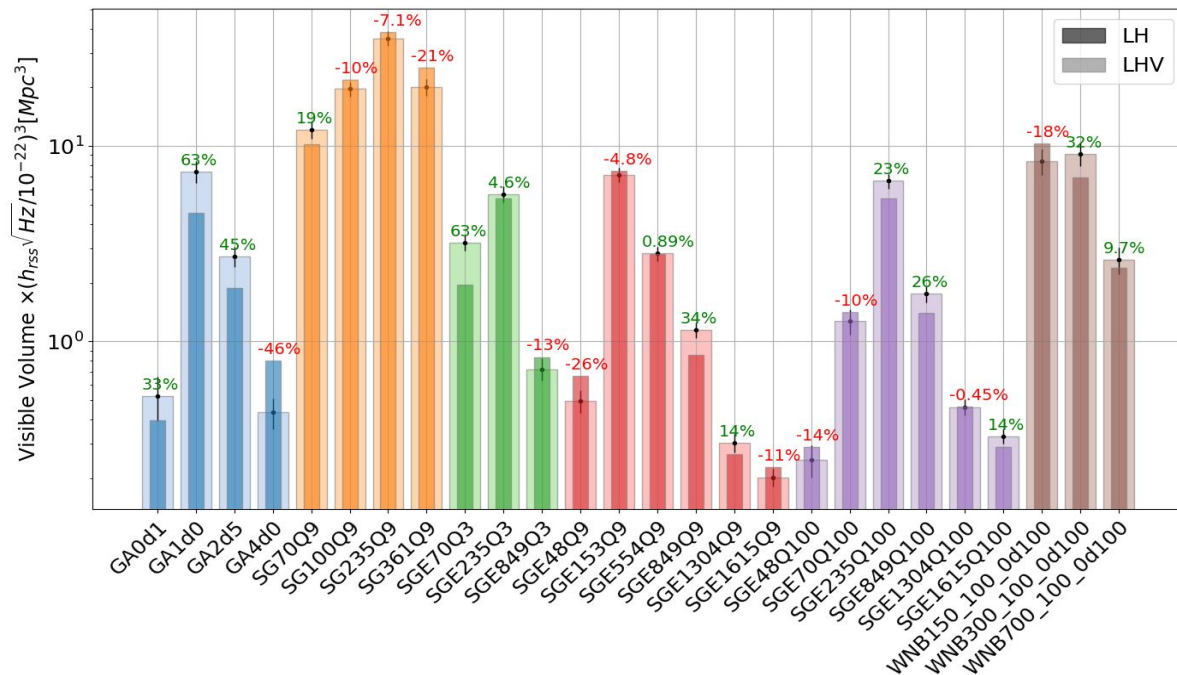
All-sky burst search on HL vs. HLV - Common and exclusive detections



Common and Exclusive detections for the two networks (“HL” - “HLV”) at FAR thresholds of 1 / y and 1 / 10 y

HL and HLV share most of the detections of louder signals

All-sky burst search on HL vs. HLV



Visible volume for short burst searches of HLV (light bars) and HL (full bars) @ IFAR = 100 years for the standard burst injection set
(injected hrss=1E-22/sqrt(Hz) @ 1 Mpc)

HL and HLV short burst searches now show comparable performances on O3b data

Conclusions - Results

- **This is the cWB-2G version used in O4 [Abac+2025]**
- **Consistency of empirical background (time-slides) with Poisson model**
- **Small correlation of False Alarms between different searches**
 - HL and HLV All-Sky searches can be treated as **independent** down to FAR threshold 1 per year, sharing ~ 20% of the reconstructed false alarms
- **Achieved performances**
 - Visible volume of HLV is **comparable** to HL's in O3 data,
 - The majority of **higher amplitude signals detections are in common** to HL and HLV
 - With respect to the O3 analysis configuration [Abbot+,Marek+] HLV now shows competitive results compared to HL

Acknowledgements

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References

- [1] Abbott, R. et al, ***All-sky search for short gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run***, *Phys. Rev. D* 104, 122004 – Published 23 December, 2021
- [2] Klimenko, S et. al. ***Method for detection and reconstruction of gravitational wave transients with networks of advanced detectors***, *Phys. Rev. D* 93, 042004, 2016
- [3] Marek J., et al. ***Search for gravitational-wave bursts in the third Advanced LIGO-Virgo run with coherent WaveBurst enhanced by machine learning***, *Phys. Rev. D* 107, 2023
- [4] Chen T, Guestric C. ***XGBoost: A Scalable Tree Boosting System***, Proceeding of ACM SIGKDD Conference, 2016
- [5] Martini et al, *in preparation*
- [6] Abac A G et al, ***All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run***, 2025, Arxiv preprint

Data set and analysis configuration

Data from O3b dataset

Net observing time 14.82 days in common to HLV detectors

January 1st, 2020 (GPS: 1262304000) - February 14, 2020 (GPS: 1265760000)

Specifics of the analysis

cWB-2G version: 6.4.6.0 **LAL version:** 7.4.1 **ROOT version:** 6.28/10

wavelet packet 10 “burst” [16, 2048 Hz] **max freq resolution** 4 Hz, **max time resolution** 1/512 s

HL analysis with “usual” hard constraint, i.e. **gamma**=-1 HLV analysis with **gamma**= -0.8

XGBoost model for Short-duration burst: trained on BKG and White Noise Bursts as for O4 analyses

XGBoost model for BBH-IMBH: trained on BKG and end of O3 Rate&Pop BBH-IMBH sets

Injections to estimate the detection efficiency

Burst standard injection set

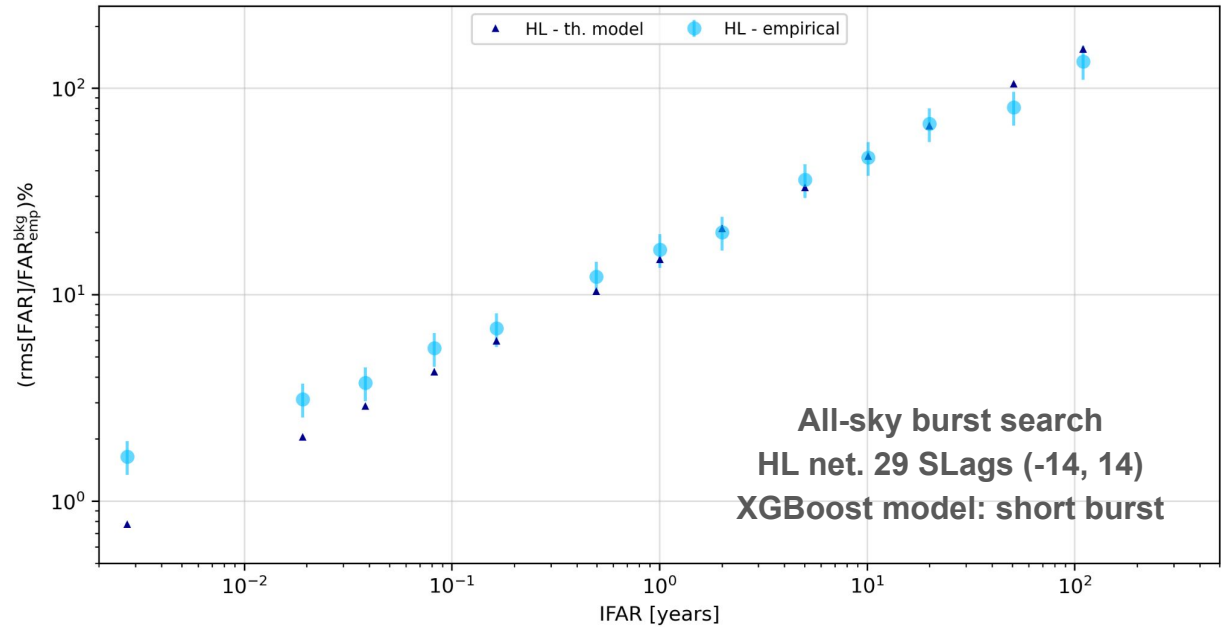
End-of-O3 Rate&Pop BBH-IMBH sets

XGBoost Features

Feature	Description
norm	sum of energies appearing in all TF resolution layers divided by energy of the event. Lower values mean the signal energy is concentrated in very few TF layers, a characteristic which is very common for background triggers.
noise	$\sqrt{\frac{1}{\sum_{\text{det}} \frac{1}{N_{\text{det}}}}}$ where N_{det} is the noise energy in each detector.
c_c	network correlation coefficient.
χ^2	noise energy per degree of freedom.
mSNR/likelihood	max value of SNR^2 in single detectors divided by the total likelihood in the network.
ecor/likelihood	sum of the off-diagonal term of likelihood matrix normalized to the total likelihood.
rho0_100d0	effective correlated SNR, capped at 100.
Qa	similarity of waveform to one dominant pulse, inspired to blip glitches.
Qp	effective number of oscillations in the waveform.

Variability of FAR estimated from time slides wrt Poisson model for HL short burst

Relative rms fluctuations of empirical FAR estimates* (dots) and Poisson predictions (triangles) at different IFAR thresholds

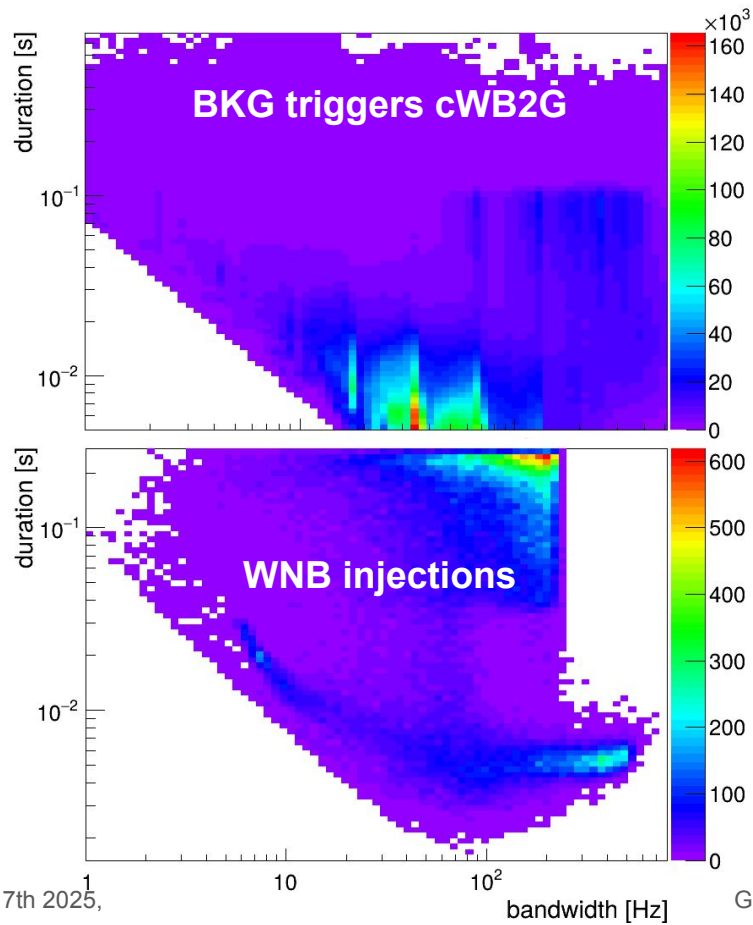


Evidence for a systematic excess of empirical FAR wrt an homogeneous Poisson point process:
sigma increases by a factor ~1.3

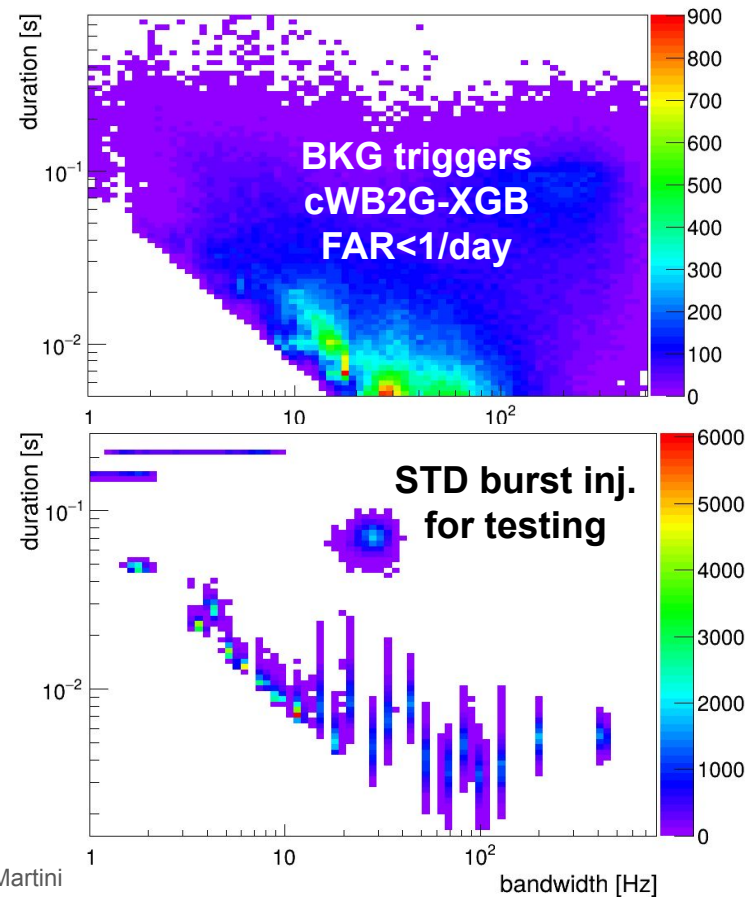
*the empirical FAR estimates are computed on different sets of time slides, shifting H by an integer number of job data segments [-14, +14]

All-sky short-burst search on HL O3 data

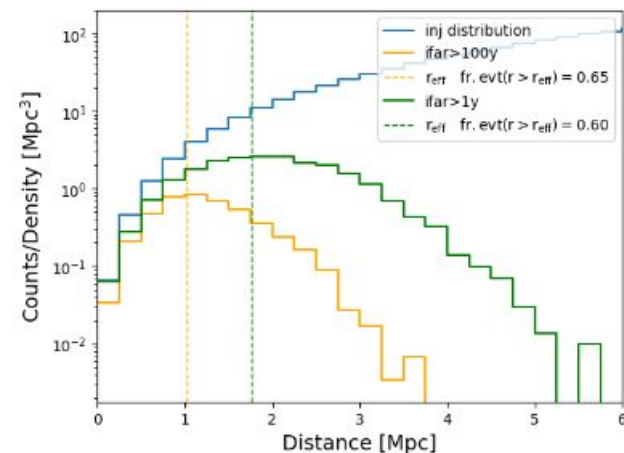
Before XGBoost post-processing



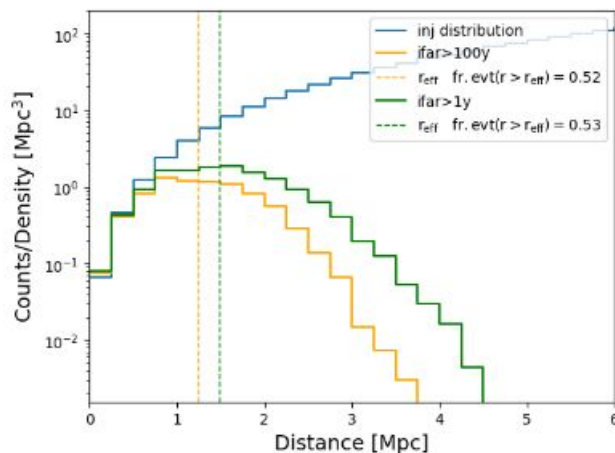
After XGBoost post-processing



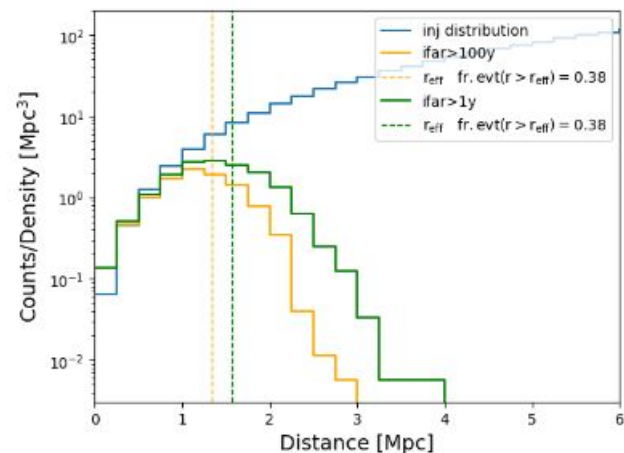
All sky burst search on HL - higher statistics, uniform in volume



(a) GA1d0



(b) SGE153Q9



(c) WNB150_100_0d1

visible volume increments as a function of distance for the HL short burst search

at IFAR thresholds = 100 years and 1 year

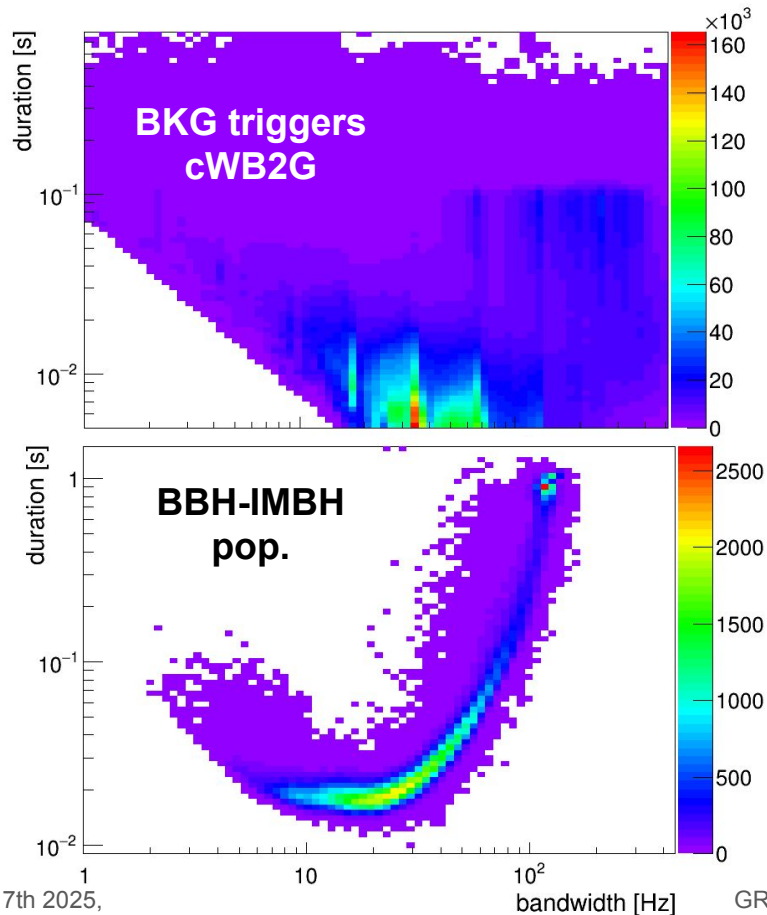
reference signal amplitude: $h_{\text{rss}} = 1\text{E-}22/\sqrt{\text{Hz}}$ at 1 Mpc

blue staircase: total volume increment contributed by distance bin

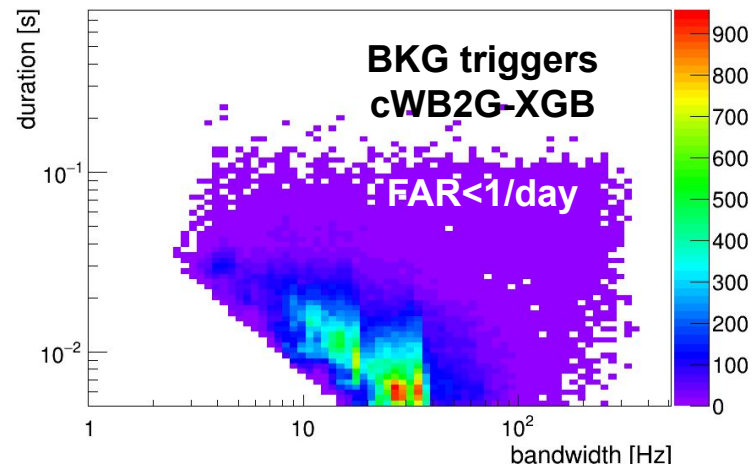
visible volume takes strong contributions by detections at large distances & low detection efficiency²⁰

All-sky short-BBH-IMBH search on HL

Before XGBoost post-processing

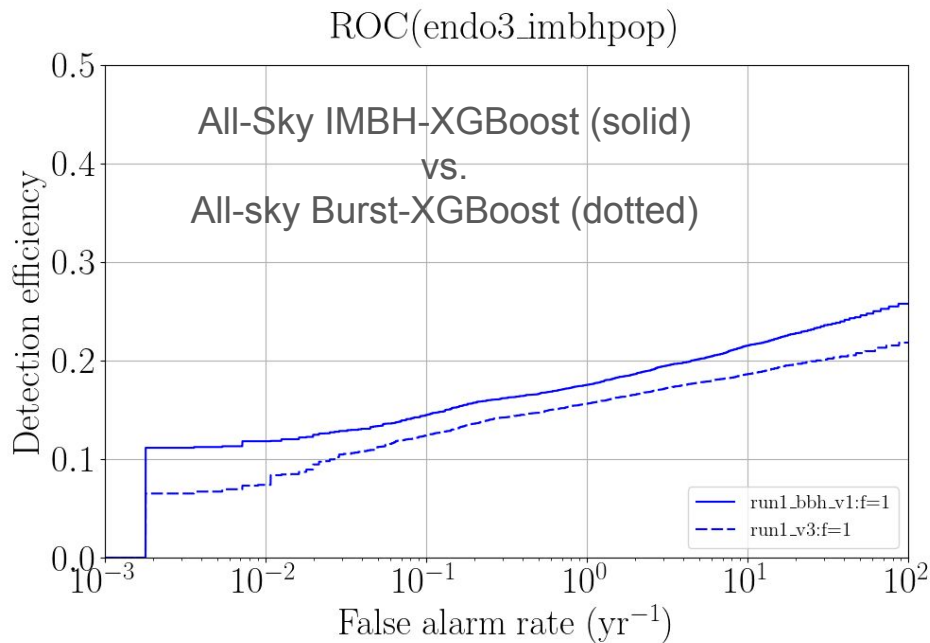
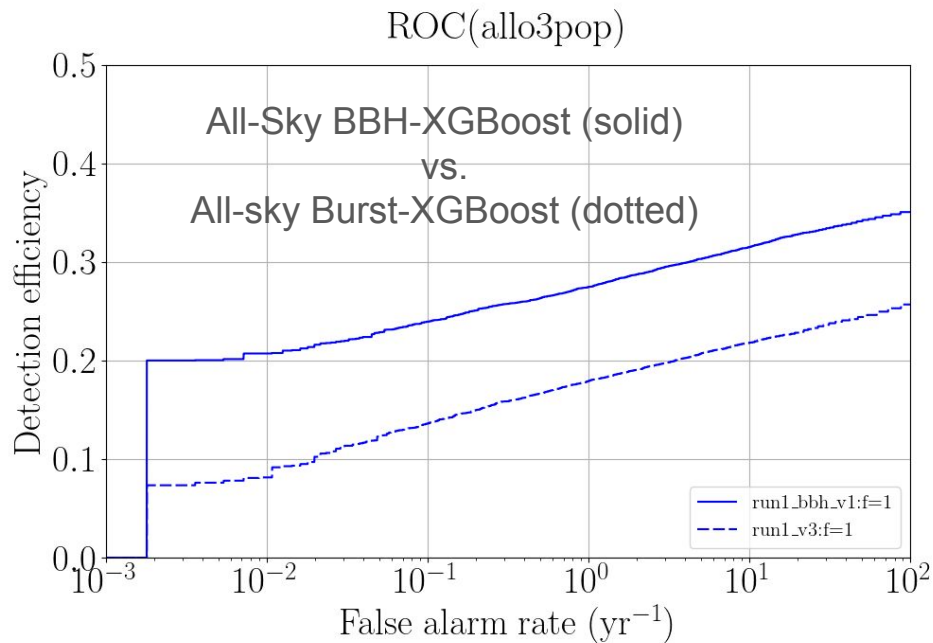


After XGBoost post-processing



testing on the end of O3
BBH-IMBH pop.

All sky burst search vs. targeted searches on HL

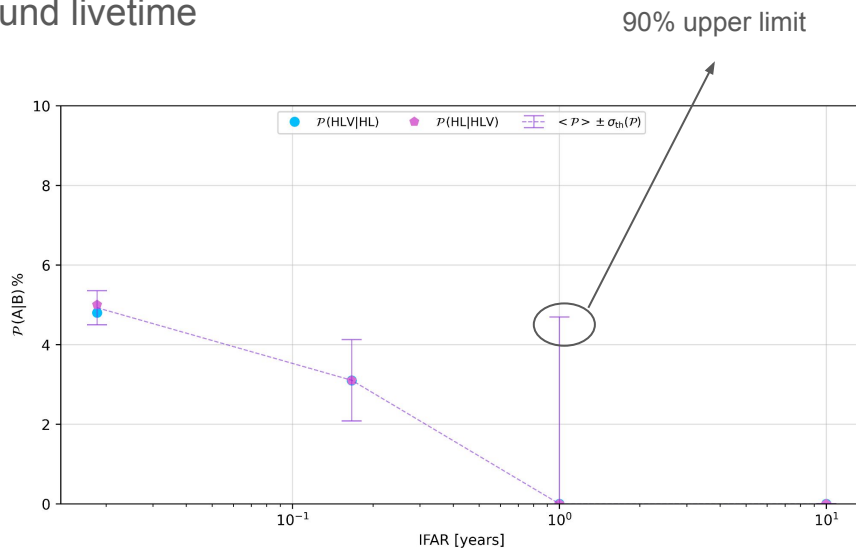
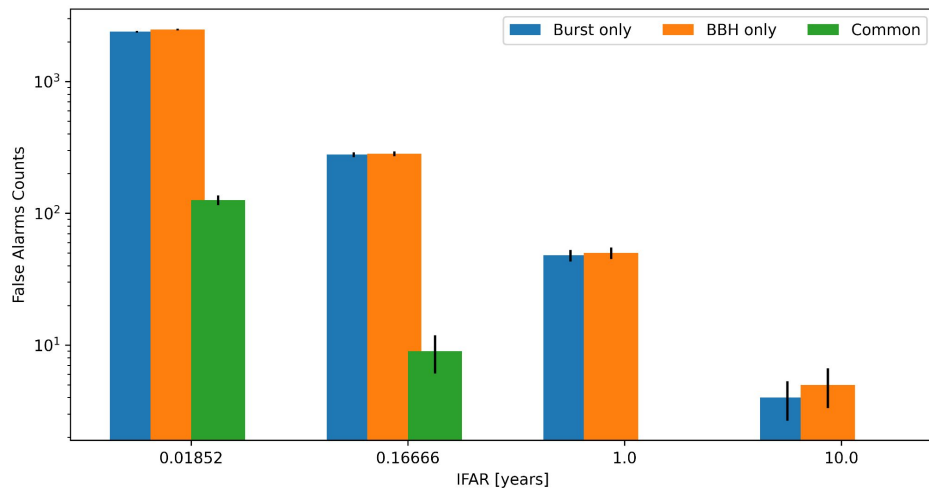


RESULTS

- **Visible gain** when **deploying a model-informed search** to target a specific astrophysical population
- **The cost** is a **complete loss of coverage** for the STD burst inj. set, apart from SGE 70Hz Q3

Correlation of false alarms between All-Sky Burst and BBH searches

- cWB2G All-sky search: **Burst vs. BBH**
- **Results based on ~48.2 years of common background livetime**



Common false alarms are only ~ 4%

- **HL short burst and BBH-IMBH targeted analyses are independent searches**